

MRAC for DCMs in the Outer Loop

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1 Introduction

Here the main steps for dealing with DCMs in the MRAC formulation are introduced. The key assumption is that $\Lambda = \lambda \mathbf{1}_m$, which holds true for the translational dynamics where $\lambda = m^{-1}$.

The outer loop can be written as

$$\dot{x}(t) = Ax(t) + B\lambda \mathbf{1}_m (u(t) + C_p(\eta_p(t))\Theta^T \Phi(t) + C_{np}(\eta_{np}(t))E_{z(t)}f(\cdot)), \quad x(t_0) = x_0, \quad t \geq t_0, \quad (1)$$

where $C_p, C_{np} : [t_0, \infty[\rightarrow \mathcal{L}(\mathbb{R}^3)$, where $\mathcal{L}(\mathbb{R}^3)$ denotes the space of *linear bounded operators* from $\mathbb{R}^3$ to $\mathbb{R}^3$. In particular $C_{(\cdot)}$ denotes a DCM given a set of planar rotations described by the angles $\eta_{(\cdot)}$.

For the sake of brevity let's delve straight into the Lyapunov analysis and isolate the elements that are needed for deriving the adaptive laws for this case scenario.

$$\begin{aligned} \dot{V}_\Theta(t, \cdot) &= 2 \left(\langle e, PB\lambda C_p(\eta_p(t))\Delta\Theta^T \Phi(t) \rangle_{\mathbb{R}^n} + \langle \Delta\Theta, \Gamma_\Theta^{-1} \dot{\Theta} \lambda \rangle_{\text{tr}} \right) \\ &= 2 \left(\langle \Phi^T(t)\Delta\Theta C_p^T(\eta_p(t))\lambda B^T P, e^T \rangle_{\text{tr}} + \langle \Delta\Theta, \Gamma_\Theta^{-1} \dot{\Theta} \lambda \rangle_{\text{tr}} \right) \\ &= 2 \left(\langle \Delta\Theta, \Phi(t)e^T PB\lambda C_p(\eta_p(t)) \rangle_{\text{tr}} + \langle \Delta\Theta, \Gamma_\Theta^{-1} \dot{\Theta} \lambda \rangle_{\text{tr}} \right) \\ &= 2 \left\langle \Delta\Theta, \left(\Gamma_\Theta^{-1} \dot{\Theta} + \Phi(t)e^T PBC_p(\eta_p(t)) \right) \lambda \right\rangle_{\text{tr}}, \end{aligned} \quad (2)$$

where the adaptive law can be easily identified as

$$\dot{\hat{\Theta}}(t) = -\Gamma_\Theta \Phi(t)e^T(t)PBC_p(\eta_p(t)), \quad \hat{\Theta}(t_0) = \hat{\Theta}_0, \quad t \geq t_0. \quad (3)$$

Similarly, we can derive an adaptive law for the nonparametric case as follows.

$$\begin{aligned} \dot{V}_f(t, \cdot) &= 2 \left(\langle e, PB\lambda C_{np}(\eta_{np}(t))E_{z(t)}\Delta f(t, \cdot) \rangle_{\mathbb{R}^n} + \langle \Delta f, \Gamma_f^{-1} \lambda \partial_t \hat{f}(t) \rangle_{\mathcal{H}} \right) \\ &= 2 \left(\langle \lambda C_{np}^T(\eta_{np}(t))B^T P e, E_{z(t)}\Delta f(t, \cdot) \rangle_{\mathbb{R}^n} + \langle \Delta f, \Gamma_f^{-1} \lambda \partial_t \hat{f}(t) \rangle_{\mathcal{H}} \right) \\ &= 2 \left\langle \Delta f, \lambda \left(\mathcal{K}_{z(t)}(\cdot)C_{np}^T(\eta_{np}(t))B^T P e + \Gamma_f^{-1} \partial_t \hat{f}(t) \right) \right\rangle_{\mathcal{H}}, \end{aligned} \quad (4)$$

where the adaptive law for the nonparametric uncertainty term can be designed as follows

$$\frac{\partial \hat{f}(t, \cdot)}{\partial t} = -\Gamma_f \mathcal{K}_{z(t)}(\cdot) C_{\text{np}}^T(\eta_{\text{np}}(t)) B^T P e(t), \quad \hat{f}(t_0, \cdot) = \hat{f}_0(\cdot), \quad t \geq t_0. \quad (5)$$